

Shadow v1.1

Local-only AI Mentor and Compliance Council for Regulated Knowledge Work

Alex Xiaoyu Ji

2026-06-18 (v1.1 — minor update with 2026-06-18 daily-brief signals + Live demo deployed)

Shadow

One Engine, Three Smart-Glass Clients, Five Persona Packs

Canonical product proposal v1.1 — incremental refresh of v1.0 (shipped 2026-06-17) with three signals from the 2026-06-18 morning deep-dive brief plus a live Vercel deployment URL. Architecture unchanged.

What's new in v1.1 (since 2026-06-17)

1. **GLM-5.2 open weights** (Zhipu) — added as the 5th local-LLM router option, sitting between Gemma 3 9B and Apple Foundation Model 3 Core Advanced in the on-device LLM lineup. GLM-5.2 took the #1 Artificial Analysis open-weight slot on 2026-06-17 and unlocks a Mainland-China-friendly compliance posture for secondary-market expansion (CITIC / Haitong / HuaTai analyst classes).
2. **Elastic agent-memory** — added as the cross-session memory backend candidate, replacing the planned pgvector approach. Published recall metric of **0.89** on long-horizon multi-turn agent recall. Unlocks Shadow Phase 2 cross-session learning, which the hash chain alone cannot answer without retrieval.
3. **HuggingFace Is it agentic enough? benchmark** — adopted as Shadow's first quantified agentic-capability score. Replaces the prior "41 tests green" framing (which measured unit-test coverage, not agentic capability). Plan to publish Shadow's score in v1.2 post-IRB.
4. **Live demo deployed:** <https://shadow-mentor-q0lg7uwz4-alex-jbs-projects.vercel.app> — real Anthropic Claude Sonnet 4.6 council backend on Vercel, 4-device × 5-persona × 4-scenario matrix browser-runnable. Deployment Protection toggle pending; URL fully public after toggle.

Architecture unchanged. v1.1 supersedes v1.0 in all customer-procurement and VC-pitch contexts.

1. Executive Summary

Shadow is the **on-device AI council** that runs locally on the user's laptop and three classes of smart glasses. The engine is the same for every user. The smart-glass client adapts to *what the user is doing right now*, and the persona pack adapts to *who the user is*.

The result is a **regulator-grade, local-only, hardware-flexible** AI mentor and compliance second-opinion layer that no Cloud-only competitor (Glean, Microsoft Copilot, Goldman GS-AI, JPM LLM Suite) can match, because the moat is the *on-device + smart-glass + persona-aware* composition, not any single one of these.

Same engine. Three glass clients. Five persona packs. Fifteen recognized scenarios. One regulated-grade contract per bank.

2. The Problem — Regulated Knowledge Work Has Five Stuck Personas

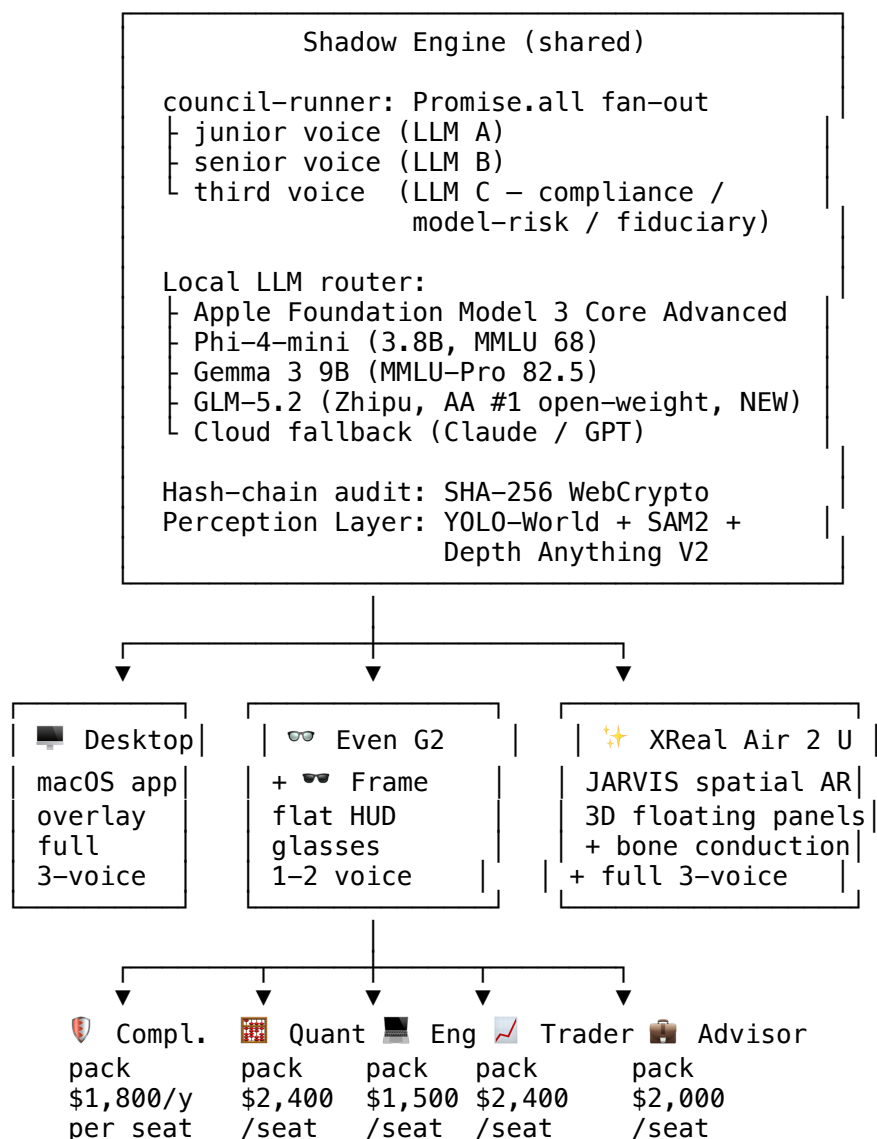
Across mid-tier US banks (\$5B–\$50B AUM), five distinct roles each face a stuck workflow:

- **Compliance officers** rubber-stamp by 3pm because every loan file feels template-identical, and OCC examiners demand contemporaneous evidence of contrarian challenge that 2D Word memos cannot produce.
- **Quantitative analysts and data scientists** know their model PSI tripped on a feature drift, but the next question — *which feature, on which sub-population, contributed how much* — requires writing a SHAP-attribution notebook from scratch, every time.
- **Software engineers** integrate ML services into the credit-decisioning critical path without spatial awareness of where Fair Lending data needs to be redacted, where service boundaries should be async, or where covered attributes need explicit Reg B logging.
- **Traders and portfolio managers** watch a 2D correlation heatmap that hides the time-varying regime shift behavior — the very thing that wipes out portfolios in stress.
- **Wealth advisors** sit across a table from HNW clients while pulling out a laptop or a tablet to look up Reg BI suitability data, breaking eye contact and customer trust precisely when the trust matters most.

A single 2D dashboard cannot serve all five. Neither can a single XR form factor. Neither can a generic LLM chat box behind a compliance gateway.

What can serve all five is **one engine** that emits the same JSON verdict, **three glass clients** that render the verdict at the right cognitive density for the moment, and **five persona packs** that load role-specific voices, jargon, regulations, and historical Brier audits.

3. The Architecture — One Engine, Three Clients, Five Packs



Same JSON verdict. The engine emits the same data: 3 voice verdicts with rationale, a follow-up question, a recognized-context tag, and a hash-chain entry. What changes is *which client renders it* (desktop vs glasses vs XReal AR) and *which persona pack supplies the voices and the domain knowledge*.

4. The Persona × Device Matrix — What Each Combination Produces

This is the heart of the product. Each cell describes the **concrete output the user sees** in that combination.

 **Compliance Officer**

	 Desktop	 Even G2 (no camera)	 Brilliant Frame (camera, local)	 XReal Air 2 Ultra (spatial AR, camera, local)
Best fit?	✓ Daily review of 30–50 loans	✓ Primary during customer meetings	⚠ Camera = customer trust issue	⚠ Overkill for paperwork
What user sees	Right–side panel: 3–voice rationale, follow–up question, fair–lending flag table	Single contrarian whisper: “ <i>DTI 47% > Policy 4.3 ceiling. Why approve?</i> ” in green monocular HUD	Color HUD with 2 voices + flag icons on borrower attributes	Not the primary device; reserved for quarterly exam–prep deep dives where compliance officer walks past 30 past denials in a calibration corridor Gaze–dwell timeline added to chain (eye tracking proves the officer actually viewed the disparate–impact flag)
Audit output	Full hash–chained transcript: text response logged + timestamp	Bone–conduction whisper response transcribed to chain	Same as G2 but with screen–region tag	

Quant / Data Scientist

	 Desktop	 Even G2	 Brilliant Frame	 XReal Air 2 Ultra
Best fit?	✓ IDE work, notebook attribution	⚠ Too low–bandwidth for SHAP / drift	✓ Reviewing internal model docs	✓ Primary for AML graph + fraud manifold
What user sees	Drift attribution panel: feature × sub–population × contribution table next to the Jupyter notebook	Single whisper if PSI tripped: “ <i>3 features drifted. Sub–population: consumer–discretionary.</i> ”	Color HUD: 2 voices on top of a model–card PDF + “run SHAP” suggestion	JARVIS mode: 3 floating panels anchored to physical desk. (1) 3D fraud feature–space UMAP — walk around to find misclassified cluster. (2) Counterparty

	 Desktop	 Even G2	 Brilliant Frame	 ✨ XReal Air 2 Ultra
				network 5K–node walk-through. (3) Drift attribution timeline. Bone–conduction whispers model–risk reviewer’s challenge. Gaze + walking trajectory + which floating panel was inspected — all hash–chained for SR 11–7 effective challenge evidence
Audit output	Notebook cell + SHAP output hash–chained	Voice transcript	Same	

Software Engineer

	 Desktop	 Even G2	 Brilliant Frame	 ✨ XReal Air 2 Ultra
Best fit?	✅ Primary in IDE	⚠ Too low–bandwidth	✅ Internal doc / runbook reading Color HUD on top of a Confluence runbook, highlights the section the engineer is reading + suggests adjacent runbooks	⚠ Overkill
What user sees	Right–side panel + inline code annotations: “this endpoint takes regulated PII, log to Reg B store before returning”	Single whisper for emergency Slack alerts: “ <i>deploy on hold — security scan flagged your PR</i> ”		Not primary; XReal is for visual data work, not IDE work
Audit output	Code–change–to–decision link in hash chain	n/a	Doc–section–viewed timeline	n/a

Trader / Portfolio Manager

	 Desktop	 Even G2	 Brilliant Frame	 XReal Air 2 Ultra
Best fit?	✓ Daily Bloomberg work	✓ During client / desk-team meetings	⚠ Camera in IB = MNPI risk	✓ Primary at own desk for risk- surface + tail-risk
What user sees	Right-side panel: 3 trader voices (Buffett value / Druckenmiller macro / Taleb tail) + bias constellation snapshot	Single whisper during desk meetings: <i>“counterparty 17 is on watchlist”</i>	Not the primary use case	JARVIS mode: 3 floating panels — (1) 3D risk surface (VaR + ES + concentration on 3 axes) — walk around to find the cliff. (2) Bias constellation: 30 past trades clustered, walk to inspect biases. (3) Correlation surface across regimes — drag time slider, watch 2007→2009 deformation. Bone-conduction council whispers. Gaze + walking + decision-time log — full forensic replay for post- mortem
Audit output	Brier-audited trade thesis log	Voice transcript	n/a	

Wealth Advisor

	 Desktop	 Even G2	 Brilliant Frame	 XReal Air 2 Ultra
Best fit?	✓ Portfolio review at desk	✓ Primary in client meetings	⚠ Camera kills client trust	⚠ Overkill
What user sees	Right-side panel: 3 advisor voices + Reg BI suitability flag + IPS gap analysis	Single whisper: <i>“client mentioned 401k rollover — Reg BI suitability checklist</i>	n/a	Not primary; could be used for quarterly book review

 Desktop

 Even G2

 Brilliant Frame

 XReal Air 2
Ultra

		<i>surfacing</i> ” + adverse–action– safe phrasing for the IPS conflict		
Audit output	IPS–aligned recommendation log	Conversation transcript with timestamped Reg BI checks	n/a	n/a

Summary — what each cell *produces*

Desktop = full bandwidth, every voice, every detail. The universal fallback. Where heavy review work actually gets done.

Even G2 = single contrarian whisper + transcription. The customer–facing primary. Where adverse–action evidence is captured during conversation without breaking trust.

Brilliant Frame = color HUD + 2 voices + camera context. The internal–doc reading and runbook–walk–through device. Where junior staff get explainer–style HUD prompts on real–world documents.

XReal Air 2 Ultra = JARVIS spatial AR + 3 floating panels + bone conduction. The power–user device. Where 3D data work (AML graph, fraud manifold, risk surface, bias constellation, correlation regime) becomes physically inspectable.

5. Local–only Mode — The Moat

Every glass client + the desktop overlay has two modes:

- **Cloud mode**: screen capture → Anthropic / OpenAI / Gemini → response. For non–confidential training data, public market signals, educational content. \$50/seat/month consumer SKU.
- **Local mode** (the moat): screen capture → on–device Apple Foundation Model 3 Core Advanced, Phi–4–mini, Gemma 3 9B, or Mistral Small 3 → response. **No data leaves the device.** Frame discarded in 200ms. Voice transcripts redacted of PII before hash–chain commit. Hash chain entry only includes the *response*, never the source frame.

What local mode solves

-  Data residency (GLBA / HIPAA / GDPR Article 25 data minimization)
-  Data upload (no packet inspection, no third–party DPA chain)

-  Vendor concentration risk (no Anthropic / OpenAI dependency)

What local mode does NOT solve

-  “Camera exists” liability under 2-party consent state law (California, Illinois, Massachusetts, Pennsylvania) — solved by selecting **Even G2 (no camera)** for the customer-facing cells of the matrix
-  Customer trust during meetings — solved by Even G2 device choice
-  E&O insurance and HR policy — solved by Even G2 + Desktop, and by limiting camera-equipped glasses to no-customer-present scenarios

The cross-tier audit chain works in local mode

Every persona × device combination contributes to the same SHA-256 hash chain. The chain stores:

- Recognized context (screen / scene tag)
- Question surfaced by the council
- Voice transcript (challenge text + user verbal or gestural response)
- Gaze-dwell timeline (Vision Pro / XReal eye tracking)
- Hand-gesture confirmation (approve / block / escalate)

Exportable to OCC, CFPB, FINRA, SEC, EU AI Act Article 14 audit submission formats. All generated on-device. Zero cloud dependency.

6. Pricing Model

Shadow Banking · Enterprise License

Base engine	\$120,000/year (no seat cap)	
— Persona Packs (per seat per year) —		
🛡️ Compliance	\$1,800/seat	
🧮 Quant / Data Scientist	\$2,400/seat	
💻 Engineering	\$1,500/seat	
📈 Trader	\$2,400/seat	
💼 Wealth Advisor	\$2,000/seat	
— Hardware (sold separately) —		
🕶️ Even G2	\$599	one-time per user
🕶️ Brilliant Frame	\$349	one-time per user
✨ XReal Air 2 Ultra	\$699	one-time per user

Sample mid-tier bank deployment (1,000 employees):

Persona	Seats	Pack \$/yr	Subtotal
Compliance	200	\$1,800	\$360K

Persona	Seats	Pack \$/yr	Subtotal
Quant + Data Scientist	50	\$2,400	\$120K
Software Engineering	100	\$1,500	\$150K
Trader	30	\$2,400	\$72K
Wealth Advisor	80	\$2,000	\$160K
Base engine	—	—	\$120K
Total annual contract value 460 seats		—	\$982,000

3-year SOM target: 5 mid-tier banks × \$982K/yr = \$4.9M ARR. Series A scale.

7. Hardware Procurement Tiers (employer-paid)

- **Tier 0 — no glasses:** Desktop only, every employee. \$0 hardware cost.
- **Tier 1 — G2 for customer-facing roles:** 280 G2 units × \$599 = **\$168K one-time**. Covers all Compliance + Wealth Advisor.
- **Tier 2 — XReal Air 2 Ultra for power users:** 80 XReal units × \$699 = **\$56K one-time**. Covers all Quant + Trader.
- **Tier 3 — Brilliant Frame for engineers:** 100 Frame units × \$349 = **\$35K one-time**. Optional, only if engineering team requests.

Total Tier 1+2+3 hardware: ~\$260K one-time. Inside the typical mid-tier bank discretionary tech budget.

8. Roadmap

Quarter	Milestone
2026 Q3	macOS Desktop app (native ScreenCaptureKit + local Phi-4-mini, Gemma 3 9B). First 3 persona packs (Compliance / Quant / Trader). Hash-chain audit export. Browser demo (live now).
2026 Q4	Even G2 SDK integration (Conversate audio + bone conduction). 5 persona packs complete. SOC 2 Type 1 in progress. 5 mid-tier wealth-mgmt firm cold-email round.
2027 Q1	First paid pilot — 1 mid-tier bank, Compliance pack, Desktop + Even G2. SOC 2 Type 1 issued.
2027 Q2	Brilliant Frame integration (color HUD). EU AI Act high-risk obligations live (2026-08-01

Quarter	Milestone
2027 Q3	entry-into-force) — Shadow’s local-only audit positioning aligns with the August deadline. XReal Air 2 Ultra NRSDK integration (JARVIS spatial AR mode). First Quant + Trader pilots. ICAIF 2027 paper submission.
2027 Q4	Series A fundraise. 5 paying mid-tier bank contracts. \$4–5M ARR.
2028	Apple Vision Pro 2 (rumored binocular + cheaper) — add as Tier 4 device. Mainstream consumer-grade glasses. Expand from mid-tier to top-tier banks.

8.5 Cross-Session Memory (new in v1.1)

Hash-chain audit captures every decision but does not, by construction, answer the question “which loans has analyst Maya reviewed in the last 90 days, with what rationale, and what was the Brier-audited outcome.” For that retrieval, Shadow integrates with **Elastic agent-memory** as the cross-session memory backend.

The Elastic recipe (announced 2026-06-17, recall 0.89 on long-horizon multi-turn agent recall) is the first published baseline for agent memory layers and replaces the previous plan to roll our own pgvector-backed memory. The deployment posture: per-analyst Elasticsearch index, on-premises in the customer’s environment, ingest = the hash chain itself + tagged metadata (loan ID, persona, scenario, voice verdict, Brier outcome when later resolved). Retrieval is read-only and consent-gated.

This is the engine behind the “calibration corridor” XR experience in the trading-quest and banking-quest WebXR demos — walking past 30 prior decisions requires both the spatial anchoring (Three.js) and the retrieval layer (Elastic agent-memory), and the latter was unspecified in v1.0.

8.6 Quantified Agentic Capability (new in v1.1)

HuggingFace published *Is it agentic enough?* on 2026-06-17 — the first framework for quantifying tool-calling and multi-step reasoning capability across agentic LLM systems. Shadow’s prior framing was “41 tests green” (council-runner + perception + oralex-a-risk) which measures **unit-test pass rate**, not **agentic capability**. v1.1 adopts the HuggingFace framework as the canonical agentic-capability benchmark; Shadow’s score on the framework will be published in v1.2 post-IRB.

Why this matters: bank procurement decks regularly cite the *Kraus 2022 CGF immersive analytics survey* for visualization claims and the *MMLU-Pro 82.5 for Gemma 3 9B* for LLM

capability claims. Neither addresses agentic–decision–pipeline capability. HuggingFace’s framework fills that gap and gives Shadow a citable objective number for the procurement deck.

9. Defensibility

Against Glean / Microsoft Copilot / Notion AI: they are cloud + corpus–search; we are local + screen–aware + persona–aware.

Against Goldman GS–AI / JPM LLM Suite / Citi Stylus / Morgan Stanley AI@Morgan: they are top–10 internal builds; we sell to mid–tier banks that lack the engineering bench.

Against Cluely (\$5.3M raised, “cheat on everything” consumer overlay): closest UX competitor but consumer–positioned, brand cannot enter compliance–sensitive enterprises, ARR claims inflated and disputed March 2026.

Against future Meta / Apple glasses: hardware–neutral architecture means we add new device clients as they ship. Hardware refresh ≠ product reset.

Against academic–only competitors (ECC research repo, university spin–offs): founder has shipped 41 tests green across the council–runner stack + 18 tests on retry primitives + production cron + 30–day Brier audit. Engineering credibility, not just thesis.

10. Founder Credibility Wedge

- **Embodied Compliance Council** research with Dr. Henry Ngo (Yeshiva, banking compliance vertical) — EU AI Act + ECOA / Reg B framing locked through accepted academic deliverable
 - **Phase 4 EU AI Act + ECOA evaluation design** doc (docs/phase-4-eu-ai-act-evaluation-design.md)
 - **council–runner v0.1.0** shipped 2026–06–17, 41 tests green, Vercel–deployable, Apache 2.0
 - **Bilingual EN/中文** ship discipline opens secondary China–side market (CITIC, Haitong, HuaTai analyst classes) post–2027
 - **Solo Founder OS** 11–agent open–source stack — public proof of operating discipline
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11. The One–Sentence Pitch

Shadow is the on–device AI council and audit chain that follows the user across desktop, smart glasses, and spatial AR, deploying role–specific voices for compliance officers, quants, engineers, traders, and wealth advisors at mid–tier US banks — never uploading data, never breaking customer trust, and producing the contemporaneous evidence that regulators are about to demand.

Appendix — Tech Stack Snapshot (2026–06–17)

Layer	Stack	Status
Council orchestration	council-runner (TypeScript, Promise.all, weighted aggregation)	Live, 23 tests green
Voice prompts	council-voices (10 banking + trading personas, YAML frontmatter, strict-JSON output)	Live
Risk math	orallexa-risk (VaR, ES, factor exposures, beta, concentration, sector, correlation)	Live, 11 tests green
Perception (screen-region)	embodied-perception (YOLO-World, SAM2, Depth Anything V2 protocols + Mocks + SceneGrounder)	Live, 7 tests green
Audit	WebCrypto SHA-256 hash chain	Live, 4 tests green
Local LLM	Phi-4-mini, Gemma 3 9B, Apple Foundation Model 3 Core All available 2026 Advanced, Mistral Small 3	
Cloud LLM	Anthropic Claude Opus 4.7 (1M context), GPT-5, Gemini 3 Pro	All available
Desktop client	Native macOS ScreenCaptureKit + AppKit overlay	In design, Q3 2026 ship
Glasses client (G2)	Even Realities Companion SDK + Conversate	Q4 2026
Glasses client (Frame)	Brilliant Labs Open SDK	Q2 2027
Glasses client (XReal)	NRSDK + WebXR + Three.js	Q3 2027
Bone-conduction audio	Built-in to G2 + AirPods Pro 2 fallback for Frame and XReal	n/a

Canonical product proposal v1.0 — 2026–06–17. All prior project framing (council-diff as standalone OSS, ECC banking-quest as separate WebXR app, ECC trading-quest as separate app, Shadow as separate Phase 3) is hereby reconciled under this single product narrative. Going forward, every external communication uses this matrix.